Do Now · retrieval

Object-oriented synoptic challenge

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is the difference between Step Into and Step Over?

Q2

What is encapsulation?

Q3

How would you store many Book objects in C#?

Q4

A Borrow method should fail on a book already on loan. Give a useful test.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Model the entity — A Book has a Title and whether it is currently on loan.

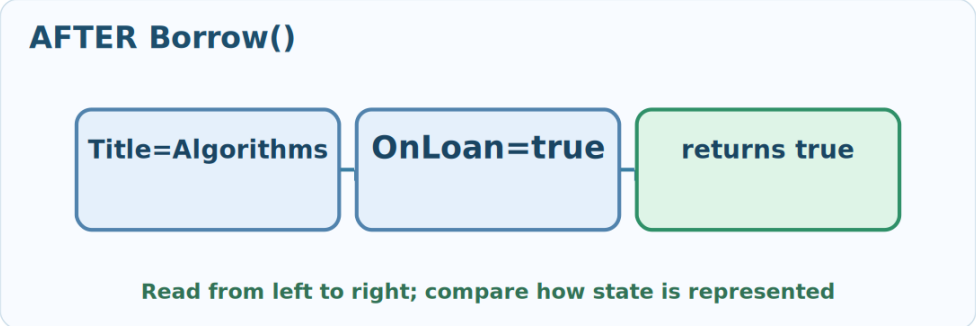
Core idea 2

Keep state protected — The public Borrow method updates a private-set OnLoan property, so external code cannot change it freely.

Core idea 3

Test a second call — The first Borrow returns true and changes state; the second call returns false because the book is already on loan.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

class — Template defining properties and methods.

object — Instance of a class.

encapsulation — Control access to and modification of object state.

collection — Stores multiple elements, such as List<Book>.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

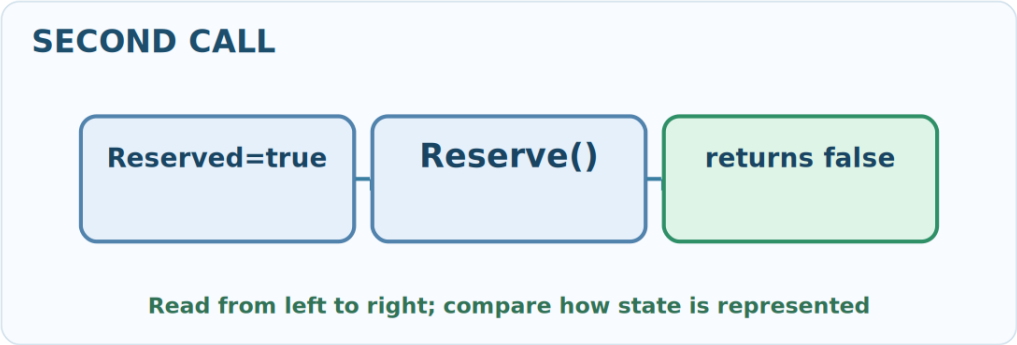
Model a Seat object. Reserve() should succeed once, then fail.

Follow the worked method

1. Predict — Initial Reserved is false, so Reserve sets it to true and returns true.

2. Model response — The second call detects Reserved already true; it returns false and leaves state unchanged.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict

The first call sets Occupied to true and returns true.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Explain the difference between a class and an object using Book.

Question 2

Write a method ReturnBook() that only succeeds if OnLoan is true, then explain one test.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain how to manage many Book objects, retrieve a book by unique ISBN, and test duplicate borrowing.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Two Book objects created from the same class must always share the same OnLoan value.

Use key vocabulary from memory

1. Explain class in your own words.

2. Explain object in your own words.

3. Explain encapsulation in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.